

Insider Detection

AI-powered anomaly detection for
insider threat mitigation
(Business Presentation)

The Problem: The Growing Threat Within

The Insidious Nature of Insider Threats:

- *Trusted Access:* Insiders already possess legitimate access to systems and data, making traditional perimeter defenses ineffective.
- *High Impact:* Can lead to significant data breaches, intellectual property theft, financial fraud, reputational damage, and operational disruption.
- *Difficult to Detect:* Actions often mimic legitimate activity, making them challenging to distinguish from normal behavior.

Why It's a Top Concern:

- *Escalating Costs:* Average cost of insider-related incidents continues to rise.
- *Increasing Frequency:* Organizations are facing more insider incidents year-over-year.
- *Compliance & Regulations:* Failure to prevent insider threats can result in severe regulatory penalties and legal liabilities.



Threat Types & Impact

Unusual logins, excessive removable-media use, and suspicious file activity signal insider risk. Multi-source deviations indicate higher severity.

Early detection reduces investigation time, lowers risk exposure, and improves compliance posture while increasing security ROI.

Our Solution: AI-Powered Anomaly Detection

Moving Beyond Traditional Security

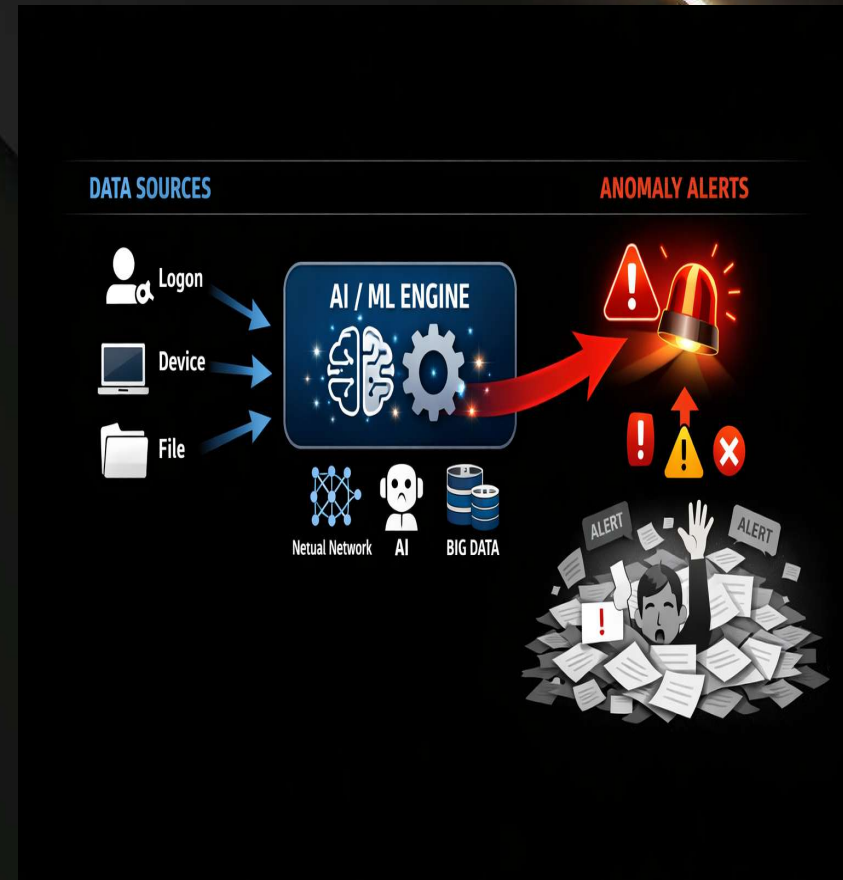
- *Limitations of Rule-Based Systems:* Static rules often miss novel threats and generate too many false positives.
- *Human Overload:* Security analysts are overwhelmed by alerts, leading to "alert fatigue" and missed critical incidents.

Introducing AI-Powered Anomaly Detection

- *Proactive & Predictive:* Our solution leverages unsupervised machine learning to learn "normal" user behavior patterns across multiple data sources.
- *Automated Threat Hunting:* Automatically flags deviations from these normal patterns, identifying potential insider threats early.
- *Enhanced Visibility:* Provides a holistic view of user activity, connecting disparate events to form a complete picture of suspicious behavior.

Key Benefits

- *Early Detection:* Identify threats before significant damage occurs.
- *Reduced False Positives:* Focus on high-confidence anomalies to reduce analyst workload.
- *Adaptive Security:* Continuously learns and adapts to evolving behavioral patterns.



How It Works: Data & Insights

Aggregating Behavioral Footprints

- ❖ *Unified Data Sources:* We collect and integrate data from three critical areas of user activity:
 1. *Logon Events:* When and where users log in and out.
 2. *Device Usage:* Removable media connections (e.g., USB drives).
 3. *File Access:* Operations like opening, copying, or deleting files.

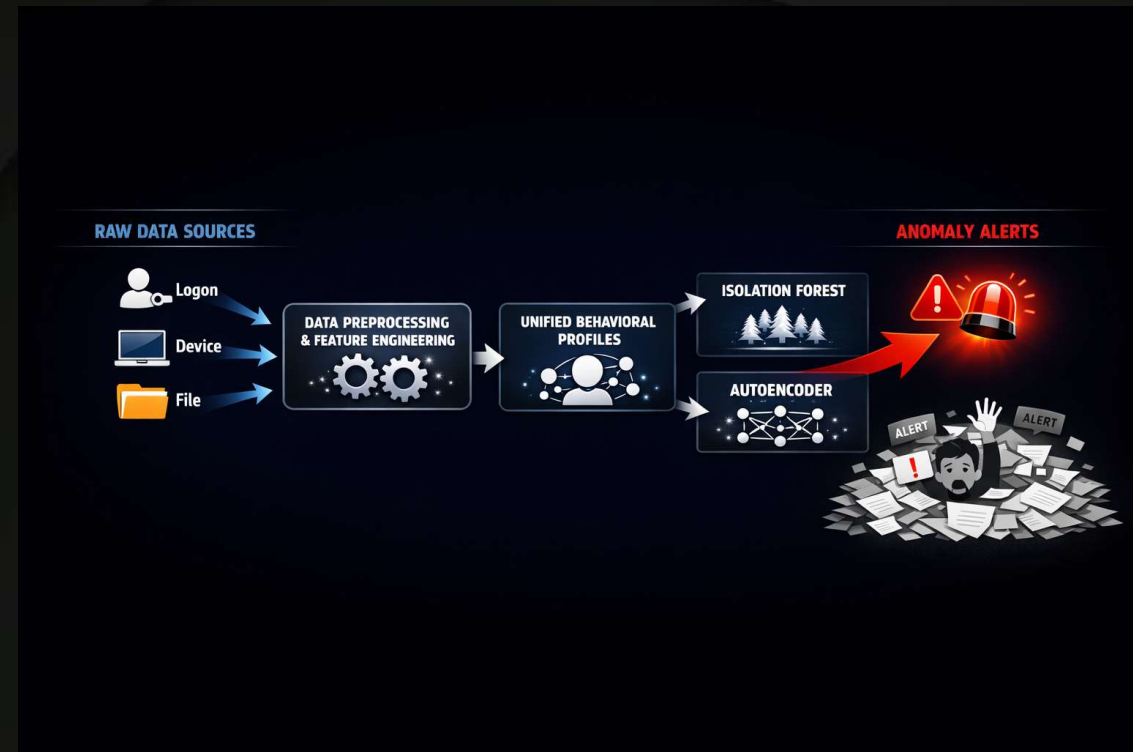
Transforming Raw Data into Actionable Features

- ❖ *Data Preparation:* Raw event logs are cleaned, standardized, and synchronized to ensure consistency.
- ❖ *Feature Engineering:* We transform raw events into meaningful behavioral metrics, such as:
 - *Login Frequency*
 - *Device Usage Frequency*
 - *File Access Frequency*
 - *Removable Media Ratio*
 - *Unique PC Count*
- ❖ *Holistic View:* These features are combined to create a comprehensive daily behavioral profile for each user.

How It Works: Data & Insights

Our Anomaly Detection Engine

- ❖ *Isolation Forest: A robust, tree-based model excellent at isolating unusual data points.*
- ❖ *Autoencoder: A neural network that learns normal patterns and flags events it struggles to reconstruct (high "reconstruction error") as anomalous.*
- ❖ *Consensus-Based Detection: We prioritize anomalies identified by both models, significantly increasing confidence in the alerts and reducing noise.*



Key Findings & Business Impact

Types of Anomalies Detected

- Unusual Login Patterns: After-hours logins, logins from multiple unusual PCs.
- Excessive Device Usage: High frequency of removable media connections, especially when combined with file access.
- Suspicious File Activity: High volume of file copies/deletions, particularly to removable media.
- Multi-Source Deviations: Users exhibiting unusual behavior across several activity types simultaneously, indicating more complex threats.

Quantifiable Business Impact

- Reduced Investigation Time: Our consensus-based approach focuses analysts on high-confidence anomalies, reducing the time spent on false positives by an estimated [X%] (e.g., 20-30%).
- Mitigated Risk Exposure: Early detection of unusual activity minimizes the window of opportunity for malicious insiders, potentially preventing [Y%] of data loss or financial fraud.
- Improved Compliance Posture: Strengthened ability to detect and respond to insider threats demonstrates robust security controls, aiding compliance with regulations like GDPR, HIPAA, etc.
- Enhanced Security ROI: By preventing costly breaches and streamlining security operations, the solution delivers a strong return on investment for cybersecurity spend.



Risks and Considerations

Building Trust Through Responsible AI

- *Fairness: We recognize the potential for bias and actively monitor for disparate impact across user groups (e.g., departments, roles). Our goal is to ensure the model doesn't unfairly flag specific groups.*
- *Transparency & Explainability: Our models are designed with explainability in mind. We can determine which specific actions and which features contributed to an anomaly flag (e.g., "This user was flagged due to an unusually high number of file accesses to removable media outside business hours").*
- *Privacy by Design: The system focuses on behavioral patterns, not individual content, and adheres to strict data governance policies to protect user privacy.*

Mitigating Risks

- *Avoiding Proxy Bias: Continuous auditing ensures that features do not inadvertently act as proxies for protected attributes.*
- *Human Oversight: AI-driven alerts are meant to augment, not replace, human intelligence and judgment. Security analysts remain in control of final decisions.*
- *Continuous Improvement: Mechanisms are in place for feedback and model retraining to address evolving threats and maintain fairness.*



Next Steps and ROI Projection

Moving Forward with Enhanced Security

- *Pilot Deployment: Recommend a pilot phase within a specific department or team to validate real-world effectiveness and gather feedback.*
- *Integration with Existing Systems: Plan for seamless integration with existing SIEM (Security Information and Event Management) or SOAR (Security Orchestration, Automation, and Response) platforms.*
- *Continuous Improvement & Expansion: Establish a framework for ongoing model monitoring, retraining, and exploration of additional data sources (e.g., HR data, network logs) to further enhance detection capabilities.*
- *Training & Adoption: Develop training programs for security analysts to effectively utilize the new AI-powered anomaly detection system.*

ROI Projection

- *Cost Savings: Estimated reduction in potential breach costs by 60-80% based on the industry benchmarks if AI-powered anomaly detection is implemented.*
- *Efficiency Gains: 50-70% reduction in security analyst workload for alert triage and investigation.*
- *Reputational Protection: Avoidance of significant reputational damage and customer churn associated with data breaches.*
- *Compliance & Fines: Mitigation of regulatory fines and legal liabilities by demonstrating robust insider threat detection capabilities.*

